

Testing Plan - Black-box Testing

Range class - expand method

As per the JavaDoc:

expand

```
public static Range expand(Range range,  
                           double lowerMargin,  
                           double upperMargin)
```

Creates a new range by adding margins to an existing range. For example: `expand(new Range(2, 6), 0.25, 0.5)` returns a range from 1 to 8.

Parameters:

`range` - the range (null not permitted).

`lowerMargin` - the lower margin (expressed as a percentage of the range length).

`upperMargin` - the upper margin (expressed as a percentage of the range length).

Returns:

The expanded range.

Throws:

`InvalidParameterException` - if null range object is passed in.

Purpose

- Create a new range from an existing one

Requirements

- Null is not permitted for the range
- Margins are expressed as a percentage
- Method returns the updated range
- Error is thrown if null is passed in

Equivalence Class Partitioning (ECT)

- Weak-normal
 - Input variables
 - Range
 - Valid range
 - Non-numeric
 - Null
 - Lowermargin
 - Valid number
 - Negative number
 - Non-numeric
 - Null
 - Uppermargin
 - Valid number
 - Negative number
 - Non-numeric

- Null
- Possibly 4 tests

Test #	Range	LowerMargin	UpperMargin	Result
1	(5, 10)	0.40	1.6	Return range from 3 to 18
2	null	0.40	1.6	Throw invalid parameter error
3	(5, 10)	a	1.6	The range should not expand, should not work
4	(5, 10)	-0.40	1.6	Return range from 7 to 18

Boundary Value Analysis (BVA)

Test #	Range	LowerMargin	UpperMargin	Result
5	(5, 5)	0.40	0.8	Range should not change

DataUtilities class - calculateColumnTotal

As per the JavaDoc:

calculateColumnTotal

```
public static double calculateColumnTotal(Values2D data,
                                         int column)
```

Returns the sum of the values in one column of the supplied data table. With invalid input, a total of zero will be returned.

Parameters:

data - the table of values (null not permitted).

column - the column index (zero-based).

Returns:

The sum of the values in the specified column.

Throws:

`InvalidParameterException` - if invalid data object is passed in.

Purpose

- Sums the values of one column

Requirements

- Null is not permitted for data

- Column index is zero-based
- Error is thrown if an invalid data is passed in
- Zero is returned with invalid data input

Decision Table

- Input variables
 - Data
 - Will use mock test to remove dependency of Values2D class
 - Mock test will initialize 2 x 2 tables to pull data from
 - column

Condition	Data	Column	Output
Test 1	(3, 4) (7, 8)	0	Returns 10
Test 2	(3, 4) (7, 8)	-1	Throw error for invalid index
Test 3	(3, 4) (7, 8)	2	Throw error for invalid index
Test 4	(3, 4) (null, 8)	0	Throw error and return 0